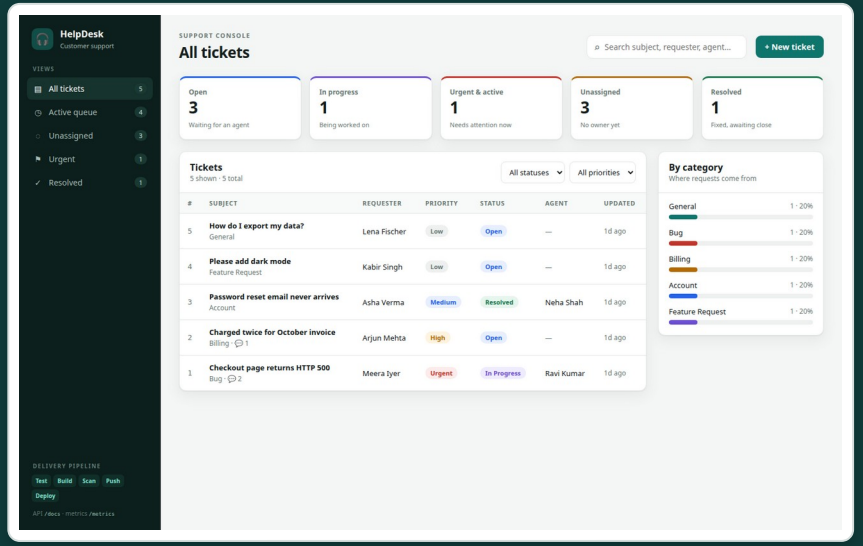
HelpDesk

A support-ticketing app, shipped with a complete DevOps pipeline on AWS

Prince Kumar
Session 21 · DevOps capstone · October 2026



M1 What HelpDesk does

1

Customers raise tickets

Bug, billing, account, feature request or general, with a priority

2

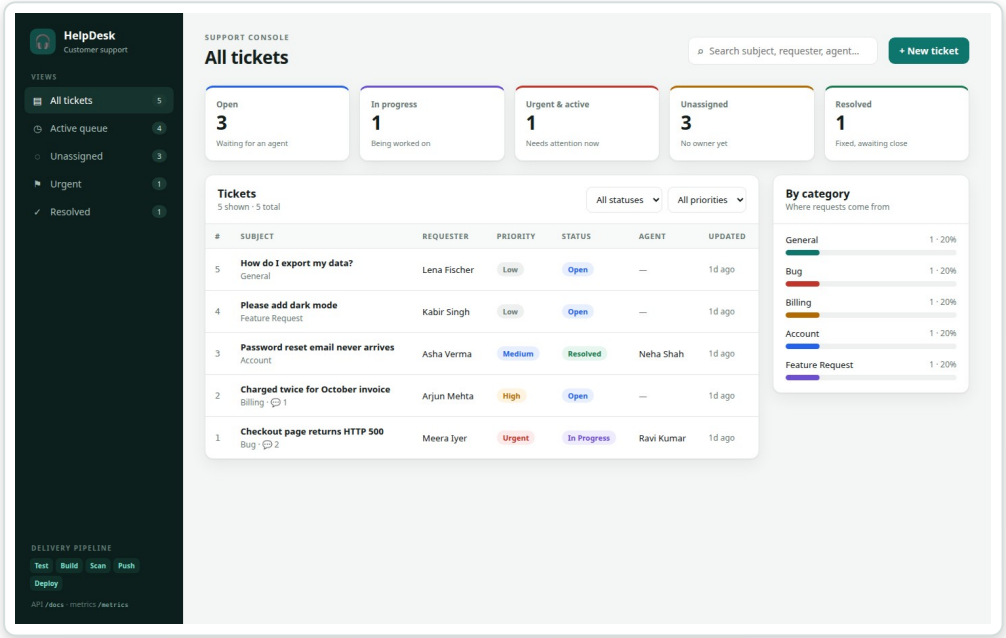
Agents work the queue

Reply to the customer, add internal notes, assign, resolve, close

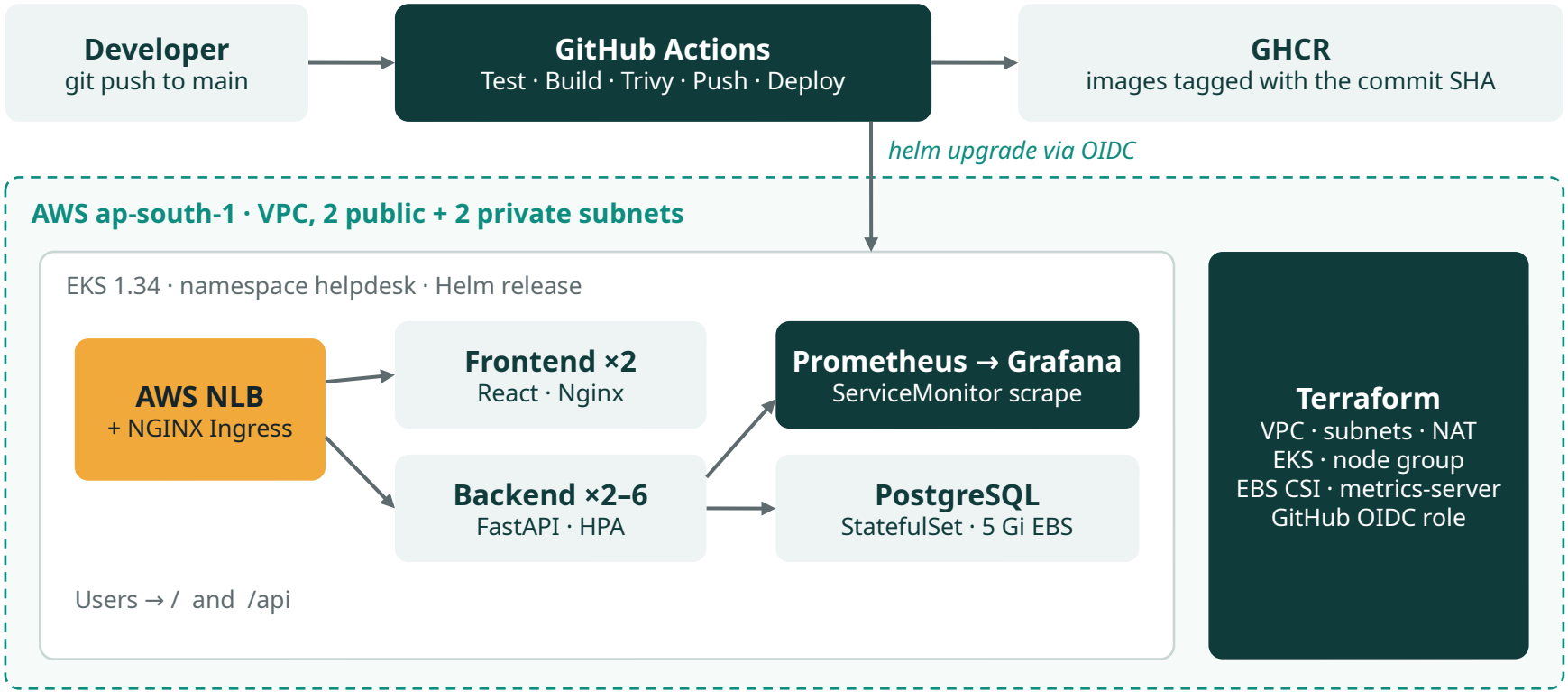
3

Leads see the big picture

Live counts: open, in progress, urgent, unassigned, by category



AR Architecture



ALL Ten modules, one pipeline

M1

Application

React · FastAPI · Postgres · Alembic

M2

Testing

pytest ×19 · ruff · SQLite test DB

M3

Git & GitHub

public repo · clear commits

M4

Docker

multi-stage · non-root · Compose

M5

CI/CD

GitHub Actions · GHCR · SHA tags

M6

DevSecOps

Trivy · fail on HIGH/CRITICAL

M7

Terraform

VPC · EKS · node group · OIDC

M8

Kubernetes

Helm · Ingress · HPA · PVC

M9

Observability

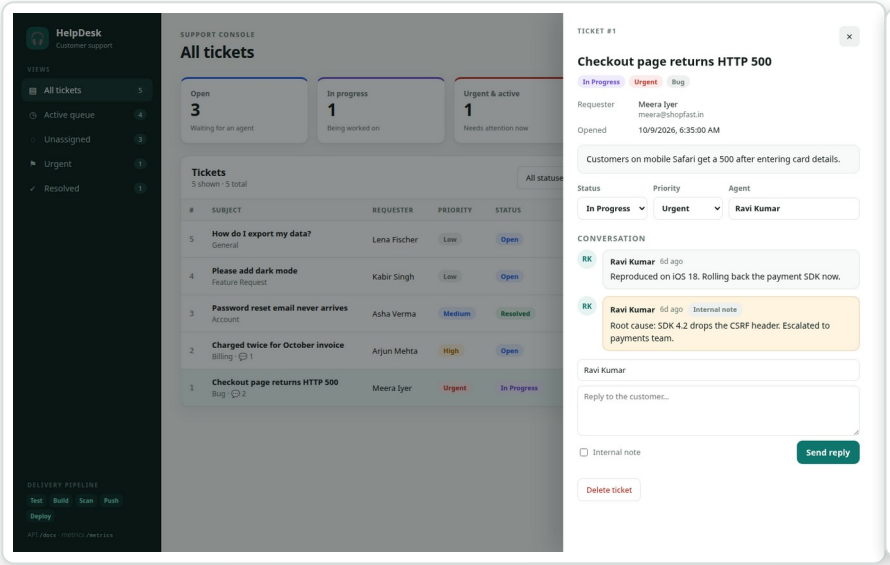
Prometheus · Grafana dashboard

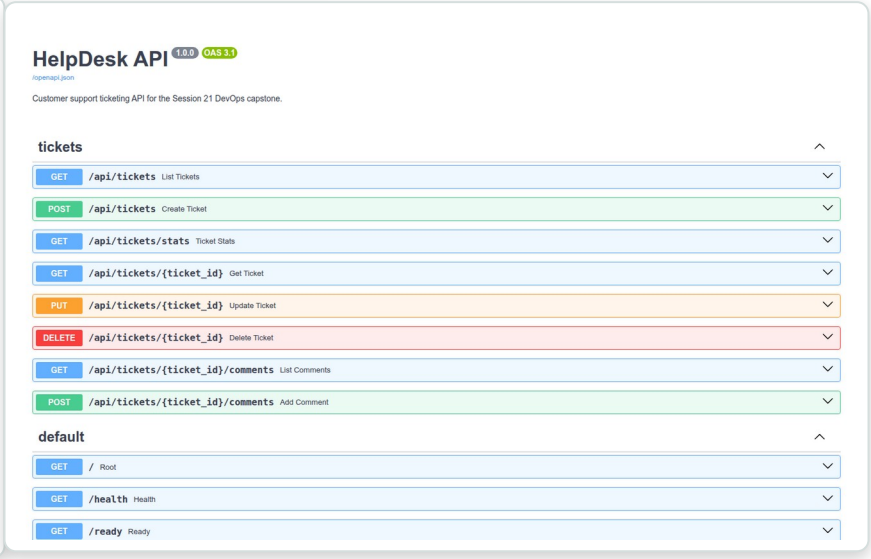
M10

Demo & docs

README · live commit → deploy

M1 Application and REST API





8

REST endpoints (CRUD + stats + comments)

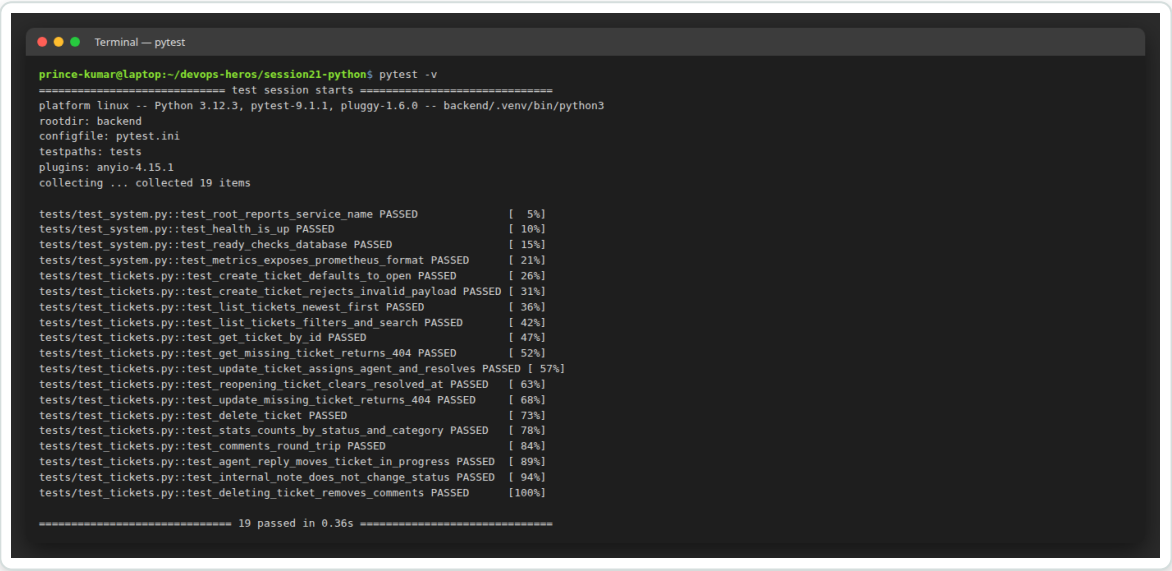
1

table schema via Alembic migration

/health · /ready · /metrics

probes for Kubernetes and Prometheus

M2 Testing and code quality



19 / 19

pytest cases passing

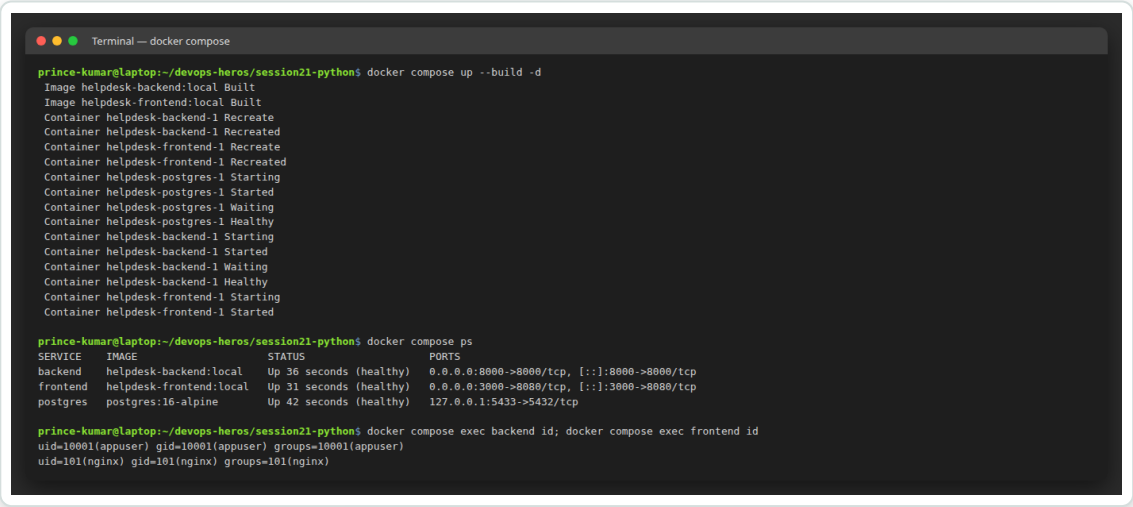
SQLite

in-memory test DB per test (conftest.py)

ruff

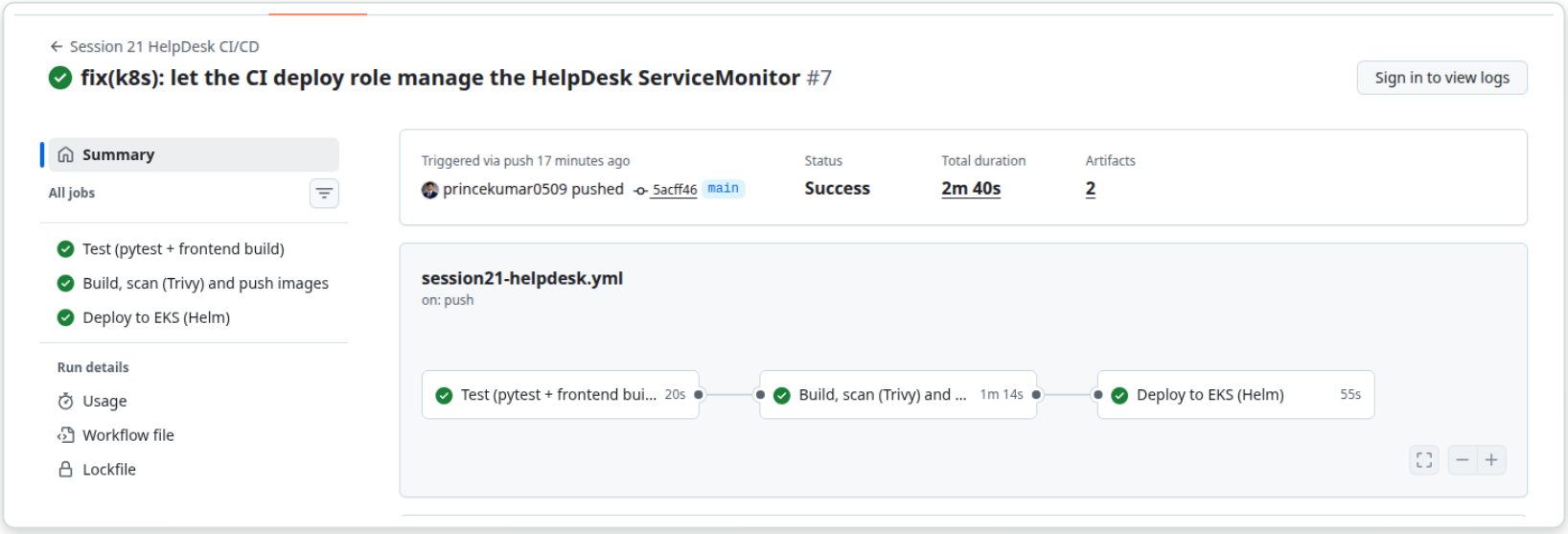
lint gate before any build

M4 Docker and Compose



- Backend: 2-stage build; pip removed from the runtime image
- Frontend: Node builds React, Nginx serves it
- Non-root: backend uid 10001, frontend uid 101
- One command starts frontend, backend and Postgres with health checks

M5 CI/CD with GitHub Actions



1 · Test
ruff · pytest · npm build · helm lint

→

2 · Build, scan, push
Docker → Trivy → GHCR :sha

→

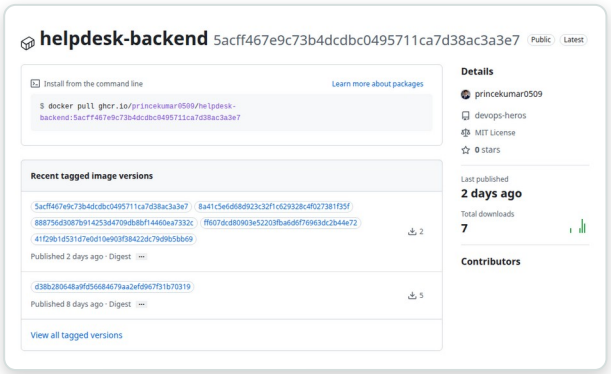
3 · Deploy
OIDC to AWS → helm upgrade on EKS

M6 Security scan and registry

Terminal — Trivy scan of the CI images from GHCR

```
prince-kumar@laptop:~/devops-heros/session21-python$ trivy image --scanners vuln,secret --severity HIGH,CRITICAL --ignore-unfix
ghcr.io/princekumar0509/helpdesk-backend:d38b280648a9…
Report Summary
```

Target	Type	Vulnerabilities	Secrets
ghcr.io/princekumar0509/helpdesk-backend:d38b280648a9fd56684679aa2efd967f31b703-19 (debian 13.7)	debian	0	-
Python	python-pkg	0	-



0

fixable HIGH / CRITICAL CVEs in either image

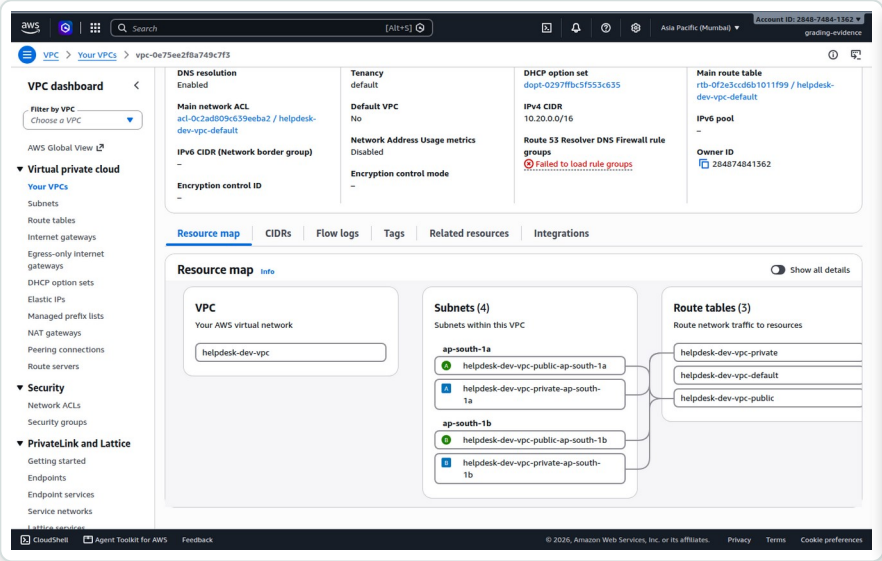
exit 1

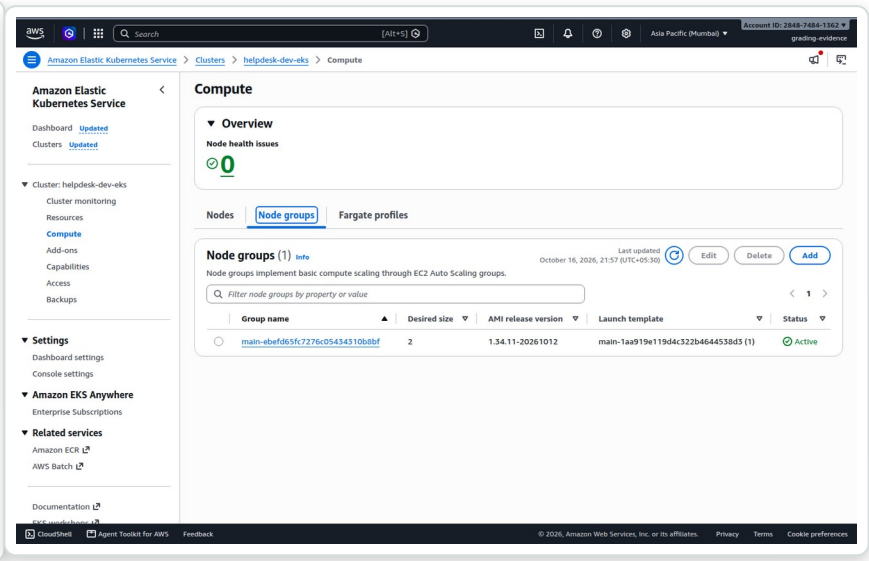
pipeline fails if Trivy finds any

:sha

every image tagged with its Git commit, never latest

M7 Terraform: VPC and EKS on AWS





65
resources in one plan

2 + 2
public + private subnets, 2 AZs

1.34
EKS, Active

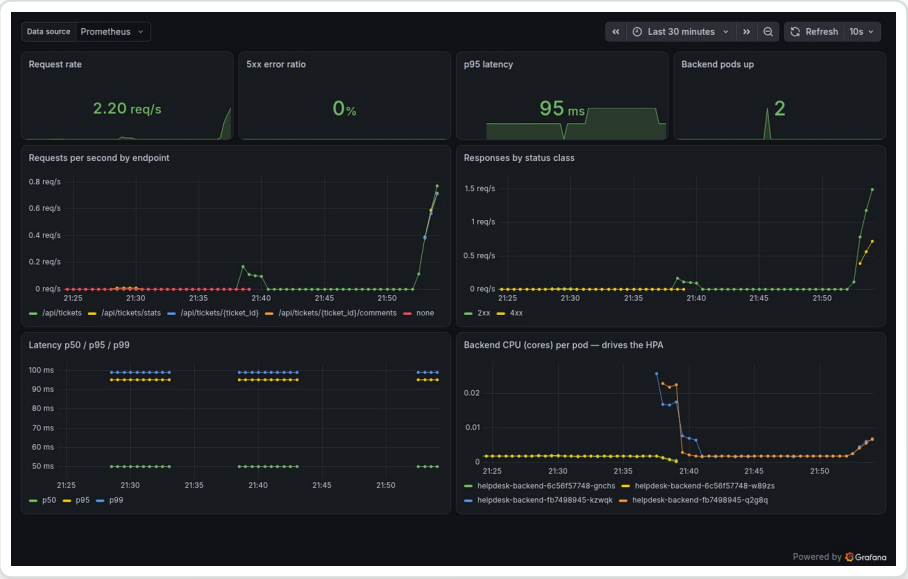
2 nodes
m7i-flex.large (free tier)

M8 Kubernetes and Helm on EKS

```
$ kubectl get pods,svc,ingress,hpa,pvc -n helpdesk
NAME                                    READY   STATUS    RESTARTS   AGE
pod/helpdesk-backend-fb7498945-kzwqk    1/1     Running   0          22m
pod/helpdesk-backend-fb7498945-q2g8q    1/1     Running   0          22m
pod/helpdesk-frontend-5f94d5b8b8-frblr  1/1     Running   0          22m
pod/helpdesk-frontend-5f94d5b8b8-lhljz  1/1     Running   0          22m
pod/helpdesk-postgres-0                 1/1     Running   0          25h
NAME                        TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)    AGE
service/helpdesk-backend    ClusterIP   172.20.83.89     <none>        8000/TCP   25h
service/helpdesk-frontend   ClusterIP   172.20.112.128   <none>        80/TCP     25h
service/helpdesk-postgres   ClusterIP   172.20.127.140   <none>        5432/TCP   25h
NAME                                CLASS   HOSTS   ADDRESS                                                                  PORTS   AGE
ingress.networking.k8s.io/helpdesk  nginx   *       afcb17c24beb84483a454772e84c72e0-39825dd2c98a88f7.elb.ap-south-1.amazonaws.com   80      25h
NAME                                                  REFERENCE                     TARGETS       MINPODS   MAXPODS   REPLICAS   AGE
horizontalpodautoscaler.autoscaling/helpdesk-backend  Deployment/helpdesk-backend   cpu: 3%/60%   2         6         2          25h
NAME                                                STATUS   VOLUME                                     CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEAT
persistentvolumeclaim/data-helpdesk-postgres-0      Bound    pvc-dfe3feaa-772e-44c2-824f-2b4b65021d34   5Gi        RWO            gp2            <unset>
```

- 2 backend + 2 frontend pods, all Running; ClusterIP Services
- Ingress: / → frontend, /api → backend, behind an AWS NLB
- Postgres StatefulSet on a 5 Gi EBS volume
- helm history: every release, one-command rollback

M9 Autoscaling and monitoring



2 / 2

Prometheus targets UP

2 → 6

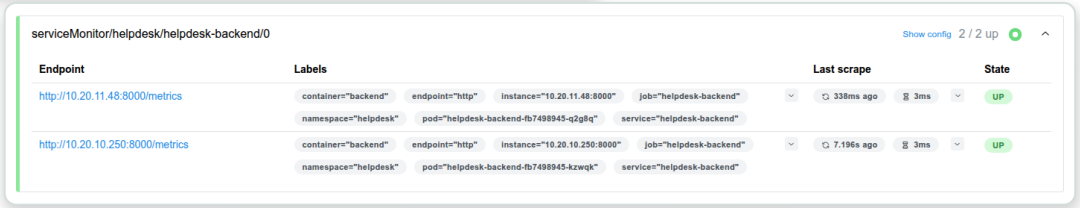
backend pods via HPA at 60% CPU

0 %

5xx error ratio

95 ms

p95 latency under load



! Real problems I solved

1 AWS account never activated

Health showed "customer verification failed". Opened a new account and verified it properly.

2 Free plan blocked t3.medium

EC2 only allows free-tier types → nodes moved to m7i-flex.large (8 GiB).

3 CI could not assume AWS role

GitHub sends an immutable OIDC subject → trust policy accepts it, set in Terraform.

4 Helm denied the ServiceMonitor

EKS namespace-admin skips CRDs → small RBAC Role, still least privilege.

Thank you

Commit → test → scan → registry → EKS, fully automated and reproducible from code

Code & docs	github.com/princekumar0509/devops-heros → session21-python
Evidence	session21-python/evidence: screenshots for every module
After grading	helm uninstall + terraform destroy, so no AWS cost is left running